



United States ENVIRONMENTAL PROTECTION AGENCY

Washington, D.C. 20460
Laboratory DMR-QA Evaluation Study 30
Laboratory Performance Evaluation
Office of Enforcement and Compliance Assurance
(These data are collected under the authority of the Federal Water Pollution Control Act.)

NPDES Permittee Data Report Form

**Due August
20, 2010**

Attention: Follow the instructions on the previous page to complete this form and submit data for evaluation.

State MI NPDES Permit Number 0020257 Permit Extension

Permittee name Cadillac WWTP

Current Permittee mailing address 200 N. Lake St

City Cadillac State MI Zip Code 49601

Phone Number (231) 775-2841 FAX Number (231) 775-2876

e-Mail lab@cadillac-mi.net

For DMR-QA Study 30, conducted in 2009, the Permittee ensured that their laboratory(s) performing the required analyses:

Received PT Samples

Yes ☒ No ☐

Submitted Complete and Accurate Data
by July 2, 2010

Yes ☒ No ☐

Received a Graded Report by
July 23, 2010

Yes ☒ No ☐

Certification by Permit Holder or Authorized Representative (as per 40 C.F.R. Section 122.22 - see instructions.)

I certify under penalty of law that this document and all attachments were prepared under my direction or supervision in accordance with a system designed to assure that qualified personnel properly gather and evaluate the information submitted. Based on my inquiry of the person or persons who manage the system, or those persons directly responsible for gathering the information, the information submitted is, to the best of my knowledge and belief, true, accurate, and complete. Each reported value was produced from a single analytical run using the analytical system that routinely performs these analyses to produce compliance monitoring data required under our National Pollutant Discharge Elimination System (NPDES) permit. Neither I nor any of my subordinates compared our results with results from independent analyses conducted by us or any other laboratory before we reported our results to the U.S.EPA. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment for knowing violations.

Name of Certifying Official Amy Vail

Title Lab Manager

Signature Amy Vail

Date 7/27/10

Address, phone number and e-mail of certifying official are required if different from above.

Address 200 N. Lake St

Phone No. (231) 775-2368

City Cadillac

State MI

Zip Code 49601

e-Mail lab@cadillac-mi.net



United States ENVIRONMENTAL PROTECTION AGENCY

Washington, D.C. 20460

Laboratory DMR-QA Evaluation Study 30

Laboratory Performance Evaluation

Office of Enforcement and Compliance Assurance

(These data are collected under the authority of the Federal Water Pollution Control Act.)

Permittee name

Cadillac WWTP

State

MI

NPDES Permit No.

0020257

Permit Extension

Identification of all CHEM, MICRO and WET laboratories who did analyses for this permit

Name of Laboratory	Address of Laboratory	U.S. EPA Lab Code	Lab Analysis Check box(es) that apply			Lab Type*
			CHEM	MICRO	WET	
Cadillac WWTP	1121 Platt Rd Cadillac, MI 49601	MI00082	☒	☒	☐	G
Kar Laboratories	4425 Manchester Rd Kalamazoo, MI 49001	MI00041	☒	☐	☐	C
Paragon Laboratories	12649 Richfield Ct Livonia, MI 48150	MI01187	☐	☐	☒	C
			☐	☐	☐	
			☐	☐	☐	
			☐	☐	☐	

*Lab Types: C = Commercial F = Federal G = Local Government I = Industrial O = Other S = State

If you need additional space, please make a copy of this page for additional laboratories.



A Waters Company

Permittee Name: Cadillac WWTP

Permit Number: MI0020257

EPA Lab Code: MI00041

Chemistry/Microbiology Analyte Checklist

DMRQA Study 30

Analyte Test / Method	Test Required	Laboratory's Graded Result	
		Acceptable	Not Acceptable (Corrective Action Required)
Minerals Alkalinity as CaCO ₃ Chloride Conductivity at 25 °C Fluoride Potassium Sodium Sulfate Total Dissolved Solids at 180 °C Total Solids at 105 °C			
Hardness Non-Filterable Residue (TSS) Calcium Magnesium Calcium Hardness as CaCO ₃ Total Hardness as CaCO ₃			
pH pH			
Settleable Solids Settleable Solids			
Solids Concentrate Non-Filterable Residue (TSS) Total Dissolved Solids at 180 °C Total Solids at 105 °C			
Solids Non-Filterable Residue (TSS) Total Dissolved Solids at 180 °C Total Solids at 105 °C			
Simple Nutrients Ammonia as N Nitrate + Nitrite as N Nitrate as N ortho-Phosphate as P			
Complex Nutrients Total Kjeldahl Nitrogen Total phosphorus as P			



A Waters Company

Permittee Name: Cadillac WWTP

Permit Number: MI0020257

EPA Lab Code: MI00041

Chemistry/Microbiology Analyte Checklist

DMRQA Study 30

Analyte Test / Method	Test Required	Laboratory's Graded Result	
		Acceptable	Not Acceptable (Corrective Action Required)
<u>Nitrite</u> Nitrite as N			
<u>Demand</u> BOD CBOD COD TOC			
<u>Oil & Grease Concentrate</u> Oil & Grease (Gravimetric)			
<u>Oil & Grease</u> Oil & Grease (Gravimetric) Oil & Grease (Infrared)			
<u>Trace Metals</u> Aluminum Antimony Arsenic Barium Beryllium Boron Cadmium Chromium Cobalt Copper Iron Lead Manganese Molybdenum Nickel Selenium Silver Strontium Thallium Vanadium Zinc			
<u>Mercury</u> Mercury			
<u>Low-Level Mercury</u> Low Level Mercury EPA 1631B	☐	☒	☐



A Waters Company

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Chemistry/Microbiology Analyte Checklist

DMRQA Study 30

Analyte Test / Method	Test Required	Laboratory's Graded Result	
		Acceptable	Not Acceptable (Corrective Action Required)
<u>Hexavalent Chromium</u> Hexavalent Chromium			
<u>Turbidity</u> Turbidity			
<u>Total Cyanide</u> Cyanide, total			
<u>Total Phenolics (4-AAP)</u> Phenolics, total			
<u>Total Residual Chlorine</u> Total Residual Chlorine			
<u>Low-Level Total Residual Chlorine</u> Low Level Total Residual Chlorine			
<u>WasteWatR™ Coliform MicrobE™</u> Total Coliforms (MF) Fecal Coliforms (MF) E.coli (MF) Total Coliforms (MPN) Fecal Coliforms (MPN) E.coli (MPN)			

Print Name

Signature/Title

Date

Use a separate checklist for EACH lab used



A Waters Company

Permittee Name: Cadillac WWTP

Permit Number: MI0020257

EPA Lab Code: MI00082

Chemistry/Microbiology Analyte Checklist

DMRQA Study 30

Analyte Test / Method	Test Required	Laboratory's Graded Result	
		Acceptable	Not Acceptable (Corrective Action Required)
Minerals Alkalinity as CaCO3 Chloride Conductivity at 25°C Fluoride Potassium Sodium Sulfate Total Dissolved Solids at 180°C Total Solids at 105°C			
Hardness Non-Filterable Residue (TSS) SM2540D Calcium Magnesium Calcium Hardness as CaCO3 Total Hardness as CaCO3	☐	☒	☐
pH pH SM4500H+ B	☐	☒	☐
Settleable Solids Settleable Solids			
Solids Concentrate Non-Filterable Residue (TSS) Total Dissolved Solids at 180°C Total Solids at 105°C			
Solids Non-Filterable Residue (TSS) Total Dissolved Solids at 180°C Total Solids at 105°C			
Simple Nutrients Ammonia as N SM4500NH3 D Nitrate + Nitrite as N Nitrate as N ortho-Phosphate as P	☐	☒	☐



A Waters Company

Permittee Name: Cadillac WWTP

Permit Number: MI0020257

EPA Lab Code: MI00082

Chemistry/Microbiology Analyte Checklist

DMRQA Study 30

Analyte Test / Method	Test Required	Laboratory's Graded Result	
		Acceptable	Not Acceptable (Corrective Action Required)
Complex Nutrients Total Kjeldahl Nitrogen Total phosphorus as P SM4500P E	☐	☒	☐
Nitrite Nitrite as N			
Demand BOD CBOD SM5210B COD TOC	☐	☒	☐
Oil & Grease Concentrate Oil & Grease (Gravimetric)			
Oil & Grease Oil & Grease (Gravimetric) Oil & Grease (Infrared)			
Trace Metals Aluminum Antimony Arsenic Barium Beryllium Boron Cadmium Chromium Cobalt Copper Iron Lead Manganese Molybdenum Nickel Selenium Silver Strontium Thallium Vanadium Zinc			



A Waters Company

Permittee Name: Cadillac WWTP

Permit Number: MI0020257

EPA Lab Code: MI00082

Chemistry/Microbiology Analyte Checklist

DMRQA Study 30

Analyte Test / Method	Test Required	Laboratory's Graded Result	
		Acceptable	Not Acceptable (Corrective Action Required)
<u>Mercury</u> Mercury			
<u>Low-Level Mercury</u> Low Level Mercury			
<u>Hexavalent Chromium</u> Hexavalent Chromium			
<u>Turbidity</u> Turbidity			
<u>Total Cyanide</u> Cyanide, total			
<u>Total Phenolics (4-AAP)</u> Phenolics, total			
<u>Total Residual Chlorine</u> Total Residual Chlorine			
<u>Low-Level Total Residual Chlorine</u> Low Level Total Residual Chlorine			
<u>WasteWatR™ Coliform MicrobE™</u> Total Coliforms (MF) Fecal Coliforms (MF) SM9222D m FC E.coli (MF) Total Coliforms (MPN) Fecal Coliforms (MPN) E.coli (MPN)	☐	☒	☐

Print Name

Signature/Title

Date

Use a separate checklist for EACH lab used



A Waters Company

Permittee Name: Cadillac WWTP

Permit Number: MI0020257

EPA Lab Code: MI01187

WET Organisms/Test Conditions/End Points Checklist DMRQA Study 30

Analyte Number	Organisms / Conditions	End Points	Test Required	Laboratory's Graded Result	
				Acceptable	Not Acceptable (Corrective Action Required)
Fathead minnow (Test Code 13)					
0754	48Hr., Acute, Non-Renewal, 25° C, MHSF	LC50	☐	☐	☐
Fathead minnow (Test Code 14)					
0755	48Hr., Acute, Non-Renewal, 25° C, 20% DMW	LC50	☐	☐	☐
Fathead minnow (Test Code 15)					
0808	7-day Short term Chronic, Daily Renewal, MHSF	IC25 (ON) Growth	☐	☐	☐
0810	7-day Short term Chronic, Daily Renewal, MHSF	NOEC (ON) Growth	☐	☐	☐
0756	7-day Short term Chronic, Daily Renewal, MHSF	NOEC Survival	☐	☐	☐
Fathead minnow (Test Code 16)					
0812	7-day Short term Chronic, Daily Renewal, 20% DMW	IC25 (ON) Growth	☐	☐	☐
0814	7-day Short term Chronic, Daily Renewal, 20% DMW	NOEC (ON) Growth	☐	☐	☐
0759	7-day Short term Chronic, Daily Renewal, 20% DMW	NOEC Survival	☐	☐	☐
Ceriodaphnia dubia (Test Code 19)					
0764	48Hr., Acute Renewal, 25° C, MHSF	LC50	☐	☐	☐
0764	48Hr., Acute Renewal, 25° C, MHSF	LC50	☐	☒	☐
Ceriodaphnia dubia (Test Code 20)					
0765	48Hr., Acute Renewal, 25° C, 20% DMW	LC50	☐	☐	☐
Ceriodaphnia dubia (Test Code 21)					
0767	7-day Short term Chronic, Daily Renewal, MHSF	IC25 Reproduction	☐	☐	☐
0767	7-day Short term Chronic, Daily Renewal, MHSF	IC25 Reproduction	☐	☒	☐
0768	7-day Short term Chronic, Daily Renewal, MHSF	NOEC Reproduction	☐	☐	☐
0768	7-day Short term Chronic, Daily Renewal, MHSF	NOEC Reproduction	☐	☒	☐
0766	7-day Short term Chronic, Daily Renewal, MHSF	NOEC Survival	☐	☐	☐
0766	7-day Short term Chronic, Daily Renewal, MHSF	NOEC Survival	☐	☒	☐
Ceriodaphnia dubia (Test Code 22)					
0770	7-day Short term Chronic, Daily Renewal, 20% DMW	IC25 Reproduction	☐	☐	☐
0771	7-day Short term Chronic, Daily Renewal, 20% DMW	NOEC Reproduction	☐	☐	☐
0769	7-day Short term Chronic, Daily Renewal, 20% DMW	NOEC Survival	☐	☐	☐
Daphnia magna (Test Code 32)					
0788	48Hr., Acute, Non-Renewal, 25° C, MHSF	LC50	☐	☐	☐
Daphnia pulex (Test Code 38)					
0794	48Hr., Acute, Non-Renewal, 25° C, MHSF	LC50	☐	☐	☐



A Waters Company

Permittee Name: Cadillac WWTP

Permit Number: MI0020257

EPA Lab Code: MI01187

WET Organisms/Test Conditions/End Points Checklist

DMRQA Study 30

Analyte Number	Organisms / Conditions	End Points	Test Required	Laboratory's Graded Result	
				Acceptable	Not Acceptable (Corrective Action Required)
Mysid (Test Code 42)					
0798	48Hr., Acute, Non-Renewal, 25° C, 40 FSW	LC50	☐	☐	☐
Inland Silverside (Test Code 45)					
0825	7-day Short term Chronic, Daily Renewal, 40 FSW	IC25 (ON) Growth	☐	☐	☐
0826	7-day Short term Chronic, Daily Renewal, 40 FSW	NOEC (ON) Growth	☐	☐	☐
0824	7-day Short term Chronic, Daily Renewal, 40 FSW	NOEC Survival	☐	☐	☐
Mysid (Test Code 43)					
0816	7-day Short term Chronic, Daily Renewal, 40 FSW	IC25 (ON) Growth	☐	☐	☐
0818	7-day Short term Chronic, Daily Renewal, 40 FSW	NOEC (ON) Growth	☐	☐	☐
0799	7-day Short term Chronic, Daily Renewal, 40 FSW	NOEC Survival	☐	☐	☐
Inland silverside (Test Code 44)					
0803	48Hr., Acute, Non-Renewal, 25° C, 40 FSW	LC50	☐	☐	☐
Sheepshead minnow (Test Code 46)					
0804	48Hr., Acute, Non-Renewal, 25° C, 40 FSW	LC50	☐	☐	☐
Sheepshead minnow (Test Code 47)					
0820	7-day Short term Chronic, Daily Renewal, 40 FSW	IC25 (ON) Growth	☐	☐	☐
0822	7-day Short term Chronic, Daily Renewal, 40 FSW	NOEC (ON) Growth	☐	☐	☐
0805	7-day Short term Chronic, Daily Renewal, 40 FSW	NOEC Survival	☐	☐	☐

Print Name

Signature/Title

Date

Use a separate checklist for EACH lab used



A Waters Company

Amy Vail
Cadillac WWTP
200 N Lake Street
Cadillac, MI 49601

DMR-QA 30 *Final Report*

DMR-QA Proficiency Testing

DMR-QA Study

Open Date: 03/15/10

Close Date: 07/02/10

Report Issued Date: 07/21/10



A Waters Company

July 21, 2010

Amy Vail
Cadillac WWTP
200 N Lake Street
Cadillac, MI 49601

Enclosed is your final report for ERA's DMR-QA 30 Proficiency Testing study. Your final report includes an evaluation of every result submitted by your facility to ERA. Please note that reports were sent on your behalf to both the USEPA DMR-QA office and your state or regional DMR-QA Coordinator. You are responsible for reviewing, signing and forwarding the enclosed analyte checklists to your DMRQA coordinator by August 20, 2010.

If you have any "Not Acceptable" evaluations for the DMR-QA 30 study, a letter of corrective action and an order form for the required remedial samples are attached for your convenience. If you have a "Not Acceptable" evaluation, but there is not an order form or a list of standards for your in-house or outside laboratories, ERA recommends that you contact your DMR-QA Coordinator for their corrective action requirements, if any.

Thank you for your participation in ERA's DMR-QA 30 Proficiency Testing study. If you have any questions, please contact myself or Curtis Wood, Quality Assurance Director, at 1-800-372-0122.

Sincerely,

Shawn Kassner
Proficiency Testing Manager

Jay R. McBurney
Quality Program Manager

attachments
smk



A Waters Company

DMRQA-30 Definitions & Study Discussion

Study Dates: 03/15/10 - 07/02/10

Report Issued: 07/21/10

DMRQA Study Definitions

The Reported Value is the value that the laboratory reported to ERA.

The ERA Assigned Values are compliant with the most current USEPA/NELAC FoPT tables. A parameter not added to the standard is given an Assigned Value of "0" per the guidelines contained in the USEPA's Criteria Document and NELAC standards.

The Acceptance Limits are established per the criteria contained in the most current USEPA/NELAC FoPT tables, or ERA's SOP for the Generation of Performance Acceptance Limits™ as applicable.

The Performance Evaluation:

- | | |
|----------------|---|
| Acceptable | = Reported Value falls within the Acceptance Limits. |
| Not Acceptable | = Reported Value falls outside the Acceptance Limits. |
| No Evaluation | = Reported Value cannot be evaluated. |
| Not Reported | = No Value reported. |

The Method Description is the method the laboratory reported to ERA.

DMRQA Study Discussion

ERA's DMR-QA 30 Proficiency Testing study has been reviewed by ERA senior management and certified compliant with the requirements of the USEPA's National Standards for Water Proficiency Testing Studies Criteria Document (December 1998), and the criteria contained in the most current NELAC FoPT tables.

ERA's DMR-QA 30 study standards were examined for any anomalies. A full review of all homogeneity, stability and accuracy verification data was completed. All analytical verification data for all analytes met the acceptance criteria contained in the USEPA's National Criteria Document for Water Proficiency Testing Studies, December 1998, and the criteria contained in the most current NELAC FoPT tables.

The data submitted by participating laboratories was also examined for study anomalies. There were four anomalies observed during the statistical review of the data. If your laboratory received the Solids Concentrate sample, catalog #4030, Complex Nutrients sample, catalog #579, the Inland Silverside (Test Code 44), catalog #WET018, or the Inland Silverside (Test Code 45) sample, catalog #WET013, these anomalies are addressed on the following page.

ERA's DMR-QA 30 study reports shall not be reproduced except in their entirety and not without the permission of the participating laboratories. The report must not be used by the participating laboratories to claim product endorsement by any agency of the U. S. government.

The data contained herein are confidential and intended for your use only.

If you have any questions or concerns regarding your assessment in ERA's DMRQA Proficiency Testing program, please contact Shawn Kassner, Proficiency Testing Manager, or Curtis Wood, Director of Regulatory Affairs and Business Development, at 1-800-372-0122.





A Waters Company

Study: **DMRQA-30**

NPDES Permit #: **MI0020257**

Laboratory Name: **Cadillac WWTP**

Inorganic Results



A Waters Company

DMRQA-30 Final Complete Report

NPDES Permit #: MI0020257
Permit Holder: Amy Vail
Lab Manager
Cadillac WWTP
1121 Plett Rd.
Cadillac, MI 49601
(231) 775-2368

ERA Customer Number: C454401
Report Issued: 07/21/10
Study Dates: 03/15/10 - 07/02/10

Analyte	Performance Evaluation	Reported Value	Assigned Value	Acceptance Limits	Units	Method Description	USEPA Lab Code	Study
Hardness								
Non-Filterable Residue (TSS)	Acceptable	53.8	55.6	43.7 - 63.2	mg/L	SM2540D	MI00082	DMRQA30
pH								
pH	Acceptable	6.20	6.30	6.10 - 6.50	S.U.	SM4500H+ B	MI00082	DMRQA30
Simple Nutrients								
Ammonia as N	Acceptable	9.16	8.47	6.25 - 10.6	mg/L	SM4500NH3 D	MI00082	DMRQA30
Complex Nutrients								
Total phosphorus as P	Acceptable	3.42	3.14	2.55 - 3.78	mg/L	SM4500P E	MI00082	DMRQA30
Demand								
CBOD	Acceptable	52.30	54.5	24.4 - 84.6	mg/L	SM5210B	MI00082	DMRQA30





A Waters Company

DMRQA-30 Final Complete Report

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 Cadillac, MI 49601
 (231) 775-2368

ERA Customer Number: C454401
 Report Issued: 07/21/10
 Study Dates: 03/15/10 - 07/02/10

Analyte	Performance Evaluation	Reported Value	Assigned Value	Acceptance Limits	Units	Method Description	USEPA Lab Code	Study
Trace Metals								
Aluminum	Acceptable	2420.	2310	1900 - 2690	µg/L	EPA 200.8	MI01187	WP184
Aluminum	Acceptable	2420.	2310	1900 - 2690	µg/L	EPA 6020A	MI01187	WP184
Antimony	Acceptable	336.	350	240 - 424	µg/L	EPA 200.8	MI01187	WP184
Antimony	Acceptable	336.	350	240 - 424	µg/L	EPA 6020A	MI01187	WP184
Arsenic	Acceptable	630.	633	531 - 741	µg/L	EPA 200.8	MI01187	WP184
Arsenic	Acceptable	630.	633	531 - 741	µg/L	EPA 6020A	MI01187	WP184
Barium	Acceptable	608.	620	538 - 699	µg/L	EPA 200.8	MI01187	WP184
Barium	Acceptable	608.	620	538 - 699	µg/L	EPA 6020A	MI01187	WP184
Beryllium	Not Acceptable	268.	322	273 - 364	µg/L	EPA 200.8	MI01187	WP184
Beryllium	Not Acceptable	268.	322	273 - 364	µg/L	EPA 6020A	MI01187	WP184
Boron	Acceptable	1410.	1550	1260 - 1800	µg/L	EPA 200.8	MI01187	WP184
Boron	Acceptable	1410.	1550	1260 - 1800	µg/L	EPA 6020A	MI01187	WP184
Cadmium	Acceptable	199.	210	179 - 239	µg/L	EPA 200.8	MI01187	WP184
Cadmium	Acceptable	199.	210	179 - 239	µg/L	EPA 6020A	MI01187	WP184
Chromium	Acceptable	100.	101	85.8 - 116	µg/L	EPA 200.8	MI01187	WP184
Chromium	Acceptable	100.	101	85.8 - 116	µg/L	EPA 6020A	MI01187	WP184
Cobalt	Acceptable	747.	765	673 - 857	µg/L	EPA 200.8	MI01187	WP184
Cobalt	Acceptable	747.	765	673 - 857	µg/L	EPA 6020A	MI01187	WP184
Copper	Acceptable	118.	115	101 - 129	µg/L	EPA 200.8	MI01187	WP184
Copper	Acceptable	118.	115	101 - 129	µg/L	EPA 6020A	MI01187	WP184
Iron	Acceptable	1130.	1130	999 - 1280	µg/L	EPA 200.8	MI01187	WP184
Iron	Acceptable	1130.	1130	999 - 1280	µg/L	EPA 6020A	MI01187	WP184
Lead	Acceptable	192.	201	170 - 231	µg/L	EPA 200.8	MI01187	WP184
Lead	Acceptable	192.	201	170 - 231	µg/L	EPA 6020A	MI01187	WP184
Manganese	Acceptable	1070.	1100	988 - 1220	µg/L	EPA 200.8	MI01187	WP184
Manganese	Acceptable	1070.	1100	988 - 1220	µg/L	EPA 6020A	MI01187	WP184
Molybdenum	Acceptable	472.	497	422 - 567	µg/L	EPA 200.8	MI01187	WP184
Molybdenum	Acceptable	472.	497	422 - 567	µg/L	EPA 6020A	MI01187	WP184
Nickel	Acceptable	778.	777	699 - 869	µg/L	EPA 200.8	MI01187	WP184
Nickel	Acceptable	778.	777	699 - 869	µg/L	EPA 6020A	MI01187	WP184
Selenium	Acceptable	1110.	1120	891 - 1300	µg/L	EPA 200.8	MI01187	WP184
Selenium	Acceptable	1110.	1120	891 - 1300	µg/L	EPA 6020A	MI01187	WP184





A Waters Company

DMRQA-30 Final Complete Report

NPDES Permit #: MI0020257
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Lab Manager
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(231) 775-2368

ERA Customer Number: C454401
Report Issued: 07/21/10
Study Dates: 03/15/10 - 07/02/10

Analyte	Performance Evaluation	Reported Value	Assigned Value	Acceptance Limits	Units	Method Description	USEPA Lab Code	Study
Trace Metals (Continued)								
Silver	Acceptable	302.	312	268 - 358	µg/L	EPA 200.8	MI01187	WP184
Silver	Acceptable	302.	312	268 - 358	µg/L	EPA 6020A	MI01187	WP184
Strontium	Acceptable	94.1	94.7	80.1 - 109	µg/L	EPA 200.8	MI01187	WP184
Strontium	Acceptable	94.1	94.7	80.1 - 109	µg/L	EPA 6020A	MI01187	WP184
Thallium	Acceptable	393.	412	325 - 499	µg/L	EPA 200.8	MI01187	WP184
Thallium	Acceptable	393.	412	325 - 499	µg/L	EPA 6020A	MI01187	WP184
Vanadium	Acceptable	847.	917	804 - 1020	µg/L	EPA 200.8	MI01187	WP184
Vanadium	Acceptable	847.	917	804 - 1020	µg/L	EPA 6020A	MI01187	WP184
Zinc	Acceptable	391.	440	377 - 509	µg/L	EPA 200.8	MI01187	WP184
Zinc	Acceptable	391.	440	377 - 509	µg/L	EPA 6020A	MI01187	WP184

Low-Level Mercury

Low Level Mercury	Acceptable	46.2	42.0	28.6 - 55.0	ng/L	EPA 1631B	MI00041	DMRQA30
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A Waters Company

Study: **DMRQA-30**

NPDES Permit #: **MI0020257**

Laboratory Name: **Cadillac WWTP**

Microbiology Results



A Waters Company

DMRQA-30 Final Complete Report

NPDES Permit #: MI0020257
Permit Holder: Amy Vail
Lab Manager
Cadillac WWTP
1121 Plett Rd.
Cadillac, MI 49601
(231) 775-2368

ERA Customer Number: C454401
Report Issued: 07/21/10
Study Dates: 03/15/10 - 07/02/10

Analyte	Performance Evaluation	Reported Value	Assigned Value	Acceptance Limits	Units	Method Description	USEPA Lab Code	Study
WasteWatR™ Coliform MicrobE™								
Fecal Coliforms (MF)	Acceptable	156.7	180	30.0 - 1080	CFU/100mL	SM9222D m FC	MI00082	DMRQA30





A Waters Company

Study: **DMRQA-30**

NPDES Permit #: **MI0020257**

Laboratory Name: **Cadillac WWTP**

WET Results



A Waters Company

DMRQA-30 Final Complete Report

NPDES Permit #: MI0020257
Permit Holder: Amy Vail
Lab Manager
Cadillac WWTP
1121 Plett Rd.
Cadillac, MI 49601
(231) 775-2368

ERA Customer Number: C454401
Report Issued: 07/21/10
Study Dates: 03/15/10 - 07/02/10

Test End Point	Performance Evaluation	Reported Value %	Assigned Value %	Acceptance Limits %	Method Description	USEPA Lab Code	Study
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Ceriodaphnia dubia (Test Code 19)
48Hr., Acute Renewal, 25° C, MHSF
Ammonium phosphate dibasic

LC50	Acceptable	54.5	57.5	21.4 - 93.5	EPA 2002.0	MI01187	DMRQA30
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Ceriodaphnia dubia (Test Code 21)
7-day Short term Chronic, Daily Renewal, MHSF
Potassium chloride

IC25 Reproduction	Acceptable	24.4	23.4	6.43 - 40.4	EPA 1002.0	MI01187	DMRQA30
NOEC Reproduction	Acceptable	25.0	12.5	6.25 - 25.0	EPA 1002.0	MI01187	DMRQA30
NOEC Survival	Acceptable	25.0	25.0	12.5 - 50.0	EPA 1002.0	MI01187	DMRQA30





A Waters Company

Permittee Name: Cadillac WWTP

Permit Number: MI0020257

EPA Lab Code: MI01187

Chemistry/Microbiology Analyte Checklist

WP Study 184

Analyte Test / Method	Test Required	Laboratory's Graded Result	
		Acceptable	Not Acceptable (Corrective Action Required)
Minerals Alkalinity as CaCO ₃ Chloride Conductivity at 25°C Fluoride Potassium Sodium Sulfate Total Dissolved Solids at 180°C Total Solids at 105°C			
Hardness Non-Filterable Residue (TSS) Calcium Magnesium Calcium Hardness as CaCO ₃ Total Hardness as CaCO ₃			
pH pH			
Settleable Solids Settleable Solids			
Solids Concentrate Non-Filterable Residue (TSS) Total Dissolved Solids at 180°C Total Solids at 105°C			
Solids Non-Filterable Residue (TSS) Total Dissolved Solids at 180°C Total Solids at 105°C			
Simple Nutrients Ammonia as N Nitrate + Nitrite as N Nitrate as N ortho-Phosphate as P			
Complex Nutrients Total Kjeldahl Nitrogen Total phosphorus as P			



A Waters Company

Permittee Name: Cadillac WWTP

Permit Number: MI0020257

EPA Lab Code: MI01187

Chemistry/Microbiology Analyte Checklist

WP Study 184

Analyte Test / Method	Test Required	Laboratory's Graded Result	
		Acceptable	Not Acceptable (Corrective Action Required)
Nitrite Nitrite as N			
Demand BOD CBOD COD TOC			
Oil & Grease Concentrate Oil & Grease (Gravimetric)			
Oil & Grease Oil & Grease (Gravimetric) Oil & Grease (Infrared)			
Trace Metals Aluminum EPA 200.8 EPA 6020A Antimony EPA 200.8 EPA 6020A Arsenic EPA 200.8 EPA 6020A Barium EPA 200.8 EPA 6020A Beryllium EPA 200.8 EPA 6020A Boron EPA 200.8 EPA 6020A Cadmium EPA 200.8 EPA 6020A Chromium EPA 200.8 EPA 6020A	 ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐	 ☒ ☒ ☒ ☒ ☒ ☒ ☒ ☒ ☐ ☐ ☒ ☒ ☒ ☒ ☒ ☒	 ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☒ ☒ ☐ ☐ ☐ ☐ ☐ ☐



A Waters Company

Permittee Name: Cadillac WWTP

Permit Number: MI0020257

EPA Lab Code: MI01187

Chemistry/Microbiology Analyte Checklist

WP Study 184

Analyte Test / Method	Test Required	Laboratory's Graded Result	
		Acceptable	Not Acceptable (Corrective Action Required)
Cobalt			
EPA 200.8	☐	☒	☐
EPA 6020A	☐	☒	☐
Copper			
EPA 200.8	☐	☒	☐
EPA 6020A	☐	☒	☐
Iron			
EPA 200.8	☐	☒	☐
EPA 6020A	☐	☒	☐
Lead			
EPA 200.8	☐	☒	☐
EPA 6020A	☐	☒	☐
Manganese			
EPA 200.8	☐	☒	☐
EPA 6020A	☐	☒	☐
Molybdenum			
EPA 200.8	☐	☒	☐
EPA 6020A	☐	☒	☐
Nickel			
EPA 200.8	☐	☒	☐
EPA 6020A	☐	☒	☐
Selenium			
EPA 200.8	☐	☒	☐
EPA 6020A	☐	☒	☐
Silver			
EPA 200.8	☐	☒	☐
EPA 6020A	☐	☒	☐
Strontium			
EPA 200.8	☐	☒	☐
EPA 6020A	☐	☒	☐
Thallium			
EPA 200.8	☐	☒	☐
EPA 6020A	☐	☒	☐
Vanadium			
EPA 200.8	☐	☒	☐
EPA 6020A	☐	☒	☐
Zinc			
EPA 200.8	☐	☒	☐
EPA 6020A	☐	☒	☐
<u>Mercury</u>			
Mercury			



A Waters Company

Permittee Name: Cadillac WWTP

Permit Number: MI0020257

EPA Lab Code: MI01187

Chemistry/Microbiology Analyte Checklist

WP Study 184

Analyte Test / Method	Test Required	Laboratory's Graded Result	
		Acceptable	Not Acceptable (Corrective Action Required)
<u>Hexavalent Chromium</u> Hexavalent Chromium			
<u>Turbidity</u> Turbidity			
<u>Total Cyanide</u> Cyanide, total			
<u>Total Phenolics (4-AAP)</u> Phenolics, total			
<u>Total Residual Chlorine</u> Total Residual Chlorine			
<u>Low-Level Total Residual Chlorine</u> Low Level Total Residual Chlorine			
<u>WasteWatR™ Colliform MicrobE™</u> Total Coliforms (MF) Fecal Coliforms (MF) E.coli (MF) Total Coliforms (MPN) Fecal Coliforms (MPN) E.coli (MPN)			

Print Name

Date

Signature/Title

Use a separate checklist for EACH lab used



A Waters Company

DMR-QA 30 Corrective Action Guidelines

Permit Holders

Upon reviewing your enclosed DMR-QA 30 final report, you may notice that some analyte results have been evaluated as 'Not Acceptable'. For these analytes, the USEPA's DMR-QA 30 announcement letter directs permit holders to 'Negotiate any 'Not Acceptable' data for corrective action with state/regional Permit Regulatory Authority' and 'Submit a completed and signed Corrective Action Report to your DMR-QA coordinator' by September 30, 2010. This means that for all 'Not Acceptable' evaluations, the laboratory must conduct an investigation, identify the problem and implement adequate corrective actions. Your state may also require the laboratory to analyze a remedial QC or PT Sample as a demonstration that the corrective action implemented was successful. ERA has contacted all state and regional coordinators and determined the type of remedial samples that they recommend or require.

If any of your 'Not Acceptable' results originated from your in-house laboratory, an order form is enclosed listing the standards that you should order and analyze. Your in-house laboratory will need to conduct its investigation and then analyze the listed standards to ensure that the identified problem has been corrected. The investigation report and, if applicable, the results of the analysis will then need to be forwarded to your DMR-QA coordinator by September 30, 2010.

You may receive an order form for both your in-house standards that have analytes evaluated as 'Not Acceptable', as well as a list of standards that your outside or contract laboratory(ies) have with 'Not Acceptable' evaluations. If there is not an order form attached or a list of standard(s) for your in-house or outside contract laboratories, then the applicable agencies do not require analysis of remedial samples.

The standards listed in the attachments are based on general guidance given to us by the state or regional coordinators. If you have any questions about your specific corrective action requirements, please contact your state or regional coordinator directly.

If you have any questions or to place an order, please contact ERA's Customer Service Group at 1-888-372-0122.



A Waters Company

DMR-QA 30 Corrective Action

Outside Laboratory List

If any of the "Not Acceptable" evaluations on your DMR-QA 30 final report originated from an outside laboratory, we have listed the name and address of the laboratory and the type of remedial sample that they can analyze for you to meet your state's corrective action requirements. Please review your DMR-QA 30 final report to confirm the analytes that each outside laboratory is responsible for and then contact the laboratory to ensure that they are conducting a corrective action investigation and to inform them of the remedial samples that they should purchase and analyze. If you have any questions concerning the type of corrective action that you and your outside laboratories are required to perform, you may contact your state or regional DMR-QA coordinator directly.

If you have any other questions, or to place an order, please contact ERA's customer service at 1-888-372-0122.

MI01187

Paragon Laboratories, Inc.

12649 Richfield Ct

Livonia, MI 48150

Trace Metals, PT Study Standard, Catalog 586



A Waters Company

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Study: **DMRQA-30**

NPDES Permit #: **MI0020257**

Laboratory Name: **Cadillac WWTP**

Inorganic Results



A Waters Company

DMRQA-30 Final Complete Report

NPDES Permit #: MI0020257
Permit Holder: Amy Vail
Lab Manager
Cadillac WWTP
1121 Plett Rd.
Cadillac, MI 49601
(231) 775-2368

ERA Customer Number: C454401
Report Issued: 07/21/10
Study Dates: 03/15/10 - 07/02/10

Analyte	Performance Evaluation	Reported Value	Assigned Value	Acceptance Limits	Units	Method Description	USEPA Lab Code	Study
Hardness								
Non-Filterable Residue (TSS)	Acceptable	53.8	55.6	43.7 - 63.2	mg/L	SM2540D	MI00082	DMRQA30
pH								
pH	Acceptable	6.20	6.30	6.10 - 6.50	S.U.	SM4500H+ B	MI00082	DMRQA30
Simple Nutrients								
Ammonia as N	Acceptable	9.16	8.47	6.25 - 10.6	mg/L	SM4500NH3 D	MI00082	DMRQA30
Complex Nutrients								
Total phosphorus as P	Acceptable	3.42	3.14	2.55 - 3.78	mg/L	SM4500P E	MI00082	DMRQA30
Demand								
CBOD	Acceptable	52.30	54.5	24.4 - 84.6	mg/L	SM5210B	MI00082	DMRQA30





A Waters Company

DMRQA-30 Final Complete Report

NPDES Permit #: MI0020257
 Permit Holder: Amy Vail
 Lab Manager
 Cadillac WWTP
 1121 Plett Rd.
 Cadillac, MI 49601
 (231) 775-2368

ERA Customer Number: C454401
 Report Issued: 07/21/10
 Study Dates: 03/15/10 - 07/02/10

Analyte	Performance Evaluation	Reported Value	Assigned Value	Acceptance Limits	Units	Method Description	USEPA Lab Code	Study
Trace Metals								
Aluminum	Acceptable	2420.	2310	1900 - 2690	µg/L	EPA 200.8	MI01187	WP184
Aluminum	Acceptable	2420.	2310	1900 - 2690	µg/L	EPA 6020A	MI01187	WP184
Antimony	Acceptable	336.	350	240 - 424	µg/L	EPA 200.8	MI01187	WP184
Antimony	Acceptable	336.	350	240 - 424	µg/L	EPA 6020A	MI01187	WP184
Arsenic	Acceptable	630.	633	531 - 741	µg/L	EPA 200.8	MI01187	WP184
Arsenic	Acceptable	630.	633	531 - 741	µg/L	EPA 6020A	MI01187	WP184
Barium	Acceptable	608.	620	538 - 699	µg/L	EPA 200.8	MI01187	WP184
Barium	Acceptable	608.	620	538 - 699	µg/L	EPA 6020A	MI01187	WP184
Beryllium	Not Acceptable	268.	322	273 - 364	µg/L	EPA 200.8	MI01187	WP184
Beryllium	Not Acceptable	268.	322	273 - 364	µg/L	EPA 6020A	MI01187	WP184
Boron	Acceptable	1410.	1550	1260 - 1800	µg/L	EPA 200.8	MI01187	WP184
Boron	Acceptable	1410.	1550	1260 - 1800	µg/L	EPA 6020A	MI01187	WP184
Cadmium	Acceptable	199.	210	179 - 239	µg/L	EPA 200.8	MI01187	WP184
Cadmium	Acceptable	199.	210	179 - 239	µg/L	EPA 6020A	MI01187	WP184
Chromium	Acceptable	100.	101	85.8 - 116	µg/L	EPA 200.8	MI01187	WP184
Chromium	Acceptable	100.	101	85.8 - 116	µg/L	EPA 6020A	MI01187	WP184
Cobalt	Acceptable	747.	765	673 - 857	µg/L	EPA 200.8	MI01187	WP184
Cobalt	Acceptable	747.	765	673 - 857	µg/L	EPA 6020A	MI01187	WP184
Copper	Acceptable	118.	115	101 - 129	µg/L	EPA 200.8	MI01187	WP184
Copper	Acceptable	118.	115	101 - 129	µg/L	EPA 6020A	MI01187	WP184
Iron	Acceptable	1130.	1130	999 - 1280	µg/L	EPA 200.8	MI01187	WP184
Iron	Acceptable	1130.	1130	999 - 1280	µg/L	EPA 6020A	MI01187	WP184
Lead	Acceptable	192.	201	170 - 231	µg/L	EPA 200.8	MI01187	WP184
Lead	Acceptable	192.	201	170 - 231	µg/L	EPA 6020A	MI01187	WP184
Manganese	Acceptable	1070.	1100	988 - 1220	µg/L	EPA 200.8	MI01187	WP184
Manganese	Acceptable	1070.	1100	988 - 1220	µg/L	EPA 6020A	MI01187	WP184
Molybdenum	Acceptable	472.	497	422 - 567	µg/L	EPA 200.8	MI01187	WP184
Molybdenum	Acceptable	472.	497	422 - 567	µg/L	EPA 6020A	MI01187	WP184
Nickel	Acceptable	778.	777	699 - 869	µg/L	EPA 200.8	MI01187	WP184
Nickel	Acceptable	778.	777	699 - 869	µg/L	EPA 6020A	MI01187	WP184
Selenium	Acceptable	1110.	1120	891 - 1300	µg/L	EPA 200.8	MI01187	WP184
Selenium	Acceptable	1110.	1120	891 - 1300	µg/L	EPA 6020A	MI01187	WP184





A Waters Company

DMRQA-30 Final Complete Report

NPDES Permit #: MI0020257
Permit Holder: Amy Vail
Lab Manager
Cadillac WWTP
1121 Plett Rd.
Cadillac, MI 49601
(231) 775-2368

ERA Customer Number: C454401
Report Issued: 07/21/10
Study Dates: 03/15/10 - 07/02/10

Analyte	Performance Evaluation	Reported Value	Assigned Value	Acceptance Limits	Units	Method Description	USEPA Lab Code	Study
Trace Metals (Continued)								
Silver	Acceptable	302.	312	268 - 358	µg/L	EPA 200.8	MI01187	WP184
Silver	Acceptable	302.	312	268 - 358	µg/L	EPA 6020A	MI01187	WP184
Strontium	Acceptable	94.1	94.7	80.1 - 109	µg/L	EPA 200.8	MI01187	WP184
Strontium	Acceptable	94.1	94.7	80.1 - 109	µg/L	EPA 6020A	MI01187	WP184
Thallium	Acceptable	393.	412	325 - 499	µg/L	EPA 200.8	MI01187	WP184
Thallium	Acceptable	393.	412	325 - 499	µg/L	EPA 6020A	MI01187	WP184
Vanadium	Acceptable	847.	917	804 - 1020	µg/L	EPA 200.8	MI01187	WP184
Vanadium	Acceptable	847.	917	804 - 1020	µg/L	EPA 6020A	MI01187	WP184
Zinc	Acceptable	391.	440	377 - 509	µg/L	EPA 200.8	MI01187	WP184
Zinc	Acceptable	391.	440	377 - 509	µg/L	EPA 6020A	MI01187	WP184

Low-Level Mercury

Low Level Mercury	Acceptable	46.2	42.0	28.6 - 55.0	ng/L	EPA 1631B	MI00041	DMRQA30
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A Waters Company

Study: **DMRQA-30**

NPDES Permit #: **MI0020257**

Laboratory Name: **Cadillac WWTP**

Microbiology Results



A Waters Company

DMRQA-30 Final Complete Report

NPDES Permit #: MI0020257
Permit Holder: Amy Vail
Lab Manager
Cadillac WWTP
1121 Plett Rd.
Cadillac, MI 49601
(231) 775-2368

ERA Customer Number: C454401
Report Issued: 07/21/10
Study Dates: 03/15/10 - 07/02/10

Analyte	Performance Evaluation	Reported Value	Assigned Value	Acceptance Limits	Units	Method Description	USEPA Lab Code	Study
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WasteWatR™ Coliform MicrobE™

Fecal Coliforms (MF)	Acceptable	156.7	180	30.0 - 1080	CFU/100mL	SM9222D m FC	MI00082	DMRQA30
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A Waters Company

WATER RESOURCES DIVISION

Cadillac
JAN 27 2011

Study: **DMRQA-30**

NPDES Permit #: **MI0020257**

Laboratory Name: **Cadillac WWTP**

WET Results



A Waters Company

DMRQA-30 Final Complete Report

NPDES Permit #: MI0020257
Permit Holder: Amy Vail
Lab Manager
Cadillac WWTP
1121 Plett Rd.
Cadillac, MI 49601
(231) 775-2368

ERA Customer Number: C454401
Report Issued: 07/21/10
Study Dates: 03/15/10 - 07/02/10

Test End Point	Performance Evaluation	Reported Value %	Assigned Value %	Acceptance Limits %	Method Description	USEPA Lab Code	Study
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Ceriodaphnia dubia (Test Code 19)
48Hr., Acute Renewal, 25° C, MHSF
Ammonium phosphate dibasic

LC50	Acceptable	54.5	57.5	21.4 - 93.5	EPA 2002.0	MI01187	DMRQA30
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Ceriodaphnia dubia (Test Code 21)
7-day Short term Chronic, Daily Renewal, MHSF
Potassium chloride

IC25 Reproduction	Acceptable	24.4	23.4	6.43 - 40.4	EPA 1002.0	MI01187	DMRQA30
NOEC Reproduction	Acceptable	25.0	12.5	6.25 - 25.0	EPA 1002.0	MI01187	DMRQA30
NOEC Survival	Acceptable	25.0	25.0	12.5 - 50.0	EPA 1002.0	MI01187	DMRQA30





A Waters Company

Study: **WP-184**

NPDES Permit #: **MI0020257**

Laboratory Name: **Cadillac WWTP**

Inorganic Results



A Waters Company

WP-184 Final Complete Report

NPDES Permit #: MI0020257
 Permit Holder: Amy Vail
 Lab Manager
 Cadillac WWTP
 1121 Plett Rd.
 Cadillac, MI 49601
 (231) 775-2368

ERA Customer Number: C454401
 Report Issued: 07/16/10
 Study Dates: 05/17/10 - 07/01/10

Analyte	Performance Evaluation	Reported Value	Assigned Value	Acceptance Limits	Units	Method Description	USEPA Lab Code	Study
Trace Metals								
Aluminum	Acceptable	2420.	2310	1900 - 2690	µg/L	EPA 200.8	MI01187	WP184
Aluminum	Acceptable	2420.	2310	1900 - 2690	µg/L	EPA 6020A	MI01187	WP184
Antimony	Acceptable	336.	350	240 - 424	µg/L	EPA 200.8	MI01187	WP184
Antimony	Acceptable	336.	350	240 - 424	µg/L	EPA 6020A	MI01187	WP184
Arsenic	Acceptable	630.	633	531 - 741	µg/L	EPA 200.8	MI01187	WP184
Arsenic	Acceptable	630.	633	531 - 741	µg/L	EPA 6020A	MI01187	WP184
Barium	Acceptable	608.	620	538 - 699	µg/L	EPA 200.8	MI01187	WP184
Barium	Acceptable	608.	620	538 - 699	µg/L	EPA 6020A	MI01187	WP184
Beryllium	Not Acceptable	268.	322	273 - 364	µg/L	EPA 200.8	MI01187	WP184
Beryllium	Not Acceptable	268.	322	273 - 364	µg/L	EPA 6020A	MI01187	WP184
Boron	Acceptable	1410.	1550	1260 - 1800	µg/L	EPA 200.8	MI01187	WP184
Boron	Acceptable	1410.	1550	1260 - 1800	µg/L	EPA 6020A	MI01187	WP184
Cadmium	Acceptable	199.	210	179 - 239	µg/L	EPA 200.8	MI01187	WP184
Cadmium	Acceptable	199.	210	179 - 239	µg/L	EPA 6020A	MI01187	WP184
Chromium	Acceptable	100.	101	85.8 - 116	µg/L	EPA 200.8	MI01187	WP184
Chromium	Acceptable	100.	101	85.8 - 116	µg/L	EPA 6020A	MI01187	WP184
Cobalt	Acceptable	747.	765	673 - 857	µg/L	EPA 200.8	MI01187	WP184
Cobalt	Acceptable	747.	765	673 - 857	µg/L	EPA 6020A	MI01187	WP184
Copper	Acceptable	118.	115	101 - 129	µg/L	EPA 200.8	MI01187	WP184
Copper	Acceptable	118.	115	101 - 129	µg/L	EPA 6020A	MI01187	WP184
Iron	Acceptable	1130.	1130	999 - 1280	µg/L	EPA 200.8	MI01187	WP184
Iron	Acceptable	1130.	1130	999 - 1280	µg/L	EPA 6020A	MI01187	WP184
Lead	Acceptable	192.	201	170 - 231	µg/L	EPA 200.8	MI01187	WP184
Lead	Acceptable	192.	201	170 - 231	µg/L	EPA 6020A	MI01187	WP184
Manganese	Acceptable	1070.	1100	988 - 1220	µg/L	EPA 200.8	MI01187	WP184
Manganese	Acceptable	1070.	1100	988 - 1220	µg/L	EPA 6020A	MI01187	WP184
Molybdenum	Acceptable	472.	497	422 - 567	µg/L	EPA 200.8	MI01187	WP184
Molybdenum	Acceptable	472.	497	422 - 567	µg/L	EPA 6020A	MI01187	WP184
Nickel	Acceptable	778.	777	699 - 869	µg/L	EPA 200.8	MI01187	WP184
Nickel	Acceptable	778.	777	699 - 869	µg/L	EPA 6020A	MI01187	WP184
Selenium	Acceptable	1110.	1120	891 - 1300	µg/L	EPA 200.8	MI01187	WP184
Selenium	Acceptable	1110.	1120	891 - 1300	µg/L	EPA 6020A	MI01187	WP184





A Waters Company

WP-184 Final Complete Report

NPDES Permit #: MI0020257
Permit Holder: Amy Vail
Lab Manager
Cadillac WWTP
1121 Plett Rd.
Cadillac, MI 49601
(231) 775-2368

ERA Customer Number: C454401
Report Issued: 07/16/10
Study Dates: 05/17/10 - 07/01/10

Analyte	Performance Evaluation	Reported Value	Assigned Value	Acceptance Limits	Units	Method Description	USEPA Lab Code	Study
Trace Metals (Continued)								
Silver	Acceptable	302.	312	268 - 358	µg/L	EPA 200.8	MI01187	WP184
Silver	Acceptable	302.	312	268 - 358	µg/L	EPA 6020A	MI01187	WP184
Strontium	Acceptable	94.1	94.7	80.1 - 109	µg/L	EPA 200.8	MI01187	WP184
Strontium	Acceptable	94.1	94.7	80.1 - 109	µg/L	EPA 6020A	MI01187	WP184
Thallium	Acceptable	393.	412	325 - 499	µg/L	EPA 200.8	MI01187	WP184
Thallium	Acceptable	393.	412	325 - 499	µg/L	EPA 6020A	MI01187	WP184
Vanadium	Acceptable	847.	917	804 - 1020	µg/L	EPA 200.8	MI01187	WP184
Vanadium	Acceptable	847.	917	804 - 1020	µg/L	EPA 6020A	MI01187	WP184
Zinc	Acceptable	391.	440	377 - 509	µg/L	EPA 200.8	MI01187	WP184
Zinc	Acceptable	391.	440	377 - 509	µg/L	EPA 6020A	MI01187	WP184





200 North Lake Street • Cadillac, Michigan 49601
231.775.0181 • fax 231.775.8755
www.cadillac-mi.net

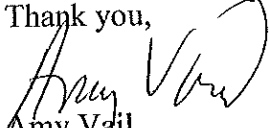
Cadillac WWTP
Westford Co
M10020257

7/27/10

Chris-

Please find our DMR-QA 30 results for this year enclosed. There is one set of results (WP 184 from Paragon Lab MI01187) that did have a "not acceptable" result but we do not use them for the analytes in that report. We only use them for WET testing. If you have any questions about this, please feel free to contact me at (231)775-2368.

Thank you,


Amy Vail
Lab Manager
City of Cadillac

WATER RESOURCES DIVISION
Cadillac
JAN 27 2011

RECEIVED
JUL 28 2010
DEQ - WATER BUREAU
GRAND RAPIDS

